

Year 10 Mathematics
Mini test 4
Quadratics



Baldivis
Secondary College

Name: Solutions

Score /30

Students will be able to use a calculator and A4 sheet of notes (single sided). This will be a 50-minute test.

Expand the following brackets and collect like term where possible: (4 Marks)

1. $x(x + 4)$ $x^2 + 4x$ ✓

2. $(x + 3)(x - 3)$ $x^2 + 3x - 3x - 9$
 $\Rightarrow x^2 - 9$ ✓

3. $(6x + 4)(2x - 8) =$ $12x^2 - 48x + 8x - 32$
 $= 12x^2 - 40x - 32$ ✓

4. $(x + 6)^2$ $(x+6)(x+6)$
 $x^2 + 6x + 6x + 36$
 $\Rightarrow x^2 + 12x + 36$ ✓

Factorise: (12 Marks)

1. $5x(x - 8) + 3(x - 8)$
 $(5x+3)(x-8)$ ✓✓

2. $x^2 + 8x + 16$
 $(x+4)(x+4)$ ✓
 $= (x+4)^2$ ✓

3. $x^2 - 16x + 63$
 $(x-9)(x-7)$
✓ ✓

4. $x^2 - 5x + 84$

$$\begin{array}{c} x \quad 12 \\ x \quad 7 \end{array} \Rightarrow (x-12)(x+7)$$

✓ ✓

5. $x^2 - 25$

$$(x-5)(x+5)$$

✓ ✓

6. $x^2 + x + 4x + 4$

$$(x+4)(x+1)$$

✓ ✓

Solve the following include all working (7 Mark)

1. $(x+4)(x+3) = 0$

$$x = -4$$

$$\text{or } x = -3$$

2. $(x-9)(x+5) = 0$

$$x = 9$$

$$\text{or } x = -5$$

3. $x^2 + 6x - 27 = 0$

$$(x+9)(x-3) = 0$$

$$x = -9$$

$$\text{or } x = 3$$

A rectangle has a width of x . The length is 5cm longer than the width.

1. Draw a diagram that represents this information. (2 marks)



2. Write an expression that represents the area of the rectangle. (1 mark)

$$A = x(x+5) \quad \checkmark$$

3. If the area of the rectangle is 36cm^2 what is the length and width? (4 Marks)

$$x(x+5) = 36 \quad \checkmark$$

$$x^2 + 5x - 36 = 0$$

$$(x+9)(x-4) = 0 \quad \checkmark$$

$$\therefore x = -9 \text{ or } x = 4 \quad \checkmark$$

Length cannot be negative

$$\therefore x = 4$$

$$\Rightarrow \text{width} = 4 \text{ cm}$$

$$\begin{aligned} \text{length} &= 4+5 \\ &= 9 \text{ cm} \end{aligned} \quad \checkmark$$

